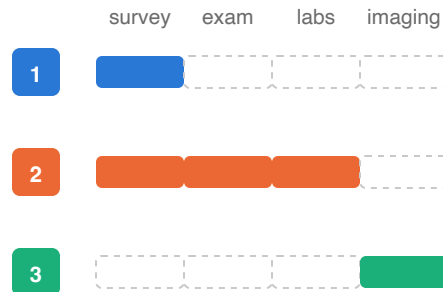


Groups with different feature spaces

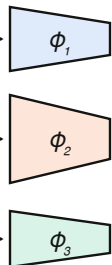
$S_g \subseteq [V]$: which measurements a group carries



dashed = not measured for that group
groups 1 and 3 share no measurement

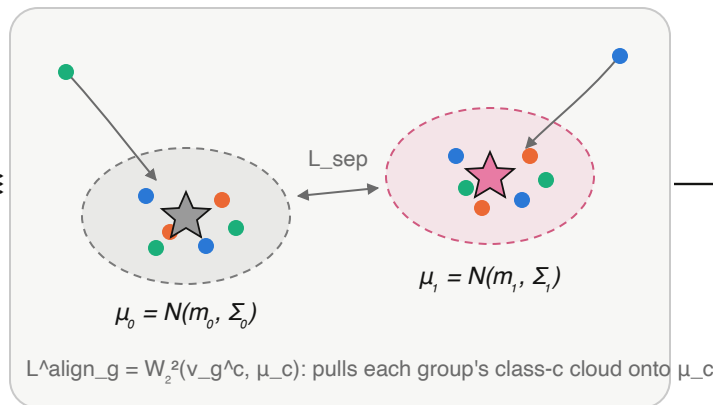
Encoders

$\phi_g : X_g \rightarrow Z$



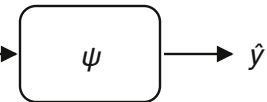
Shared latent space Z

every group is pulled onto the same class anchors



One head

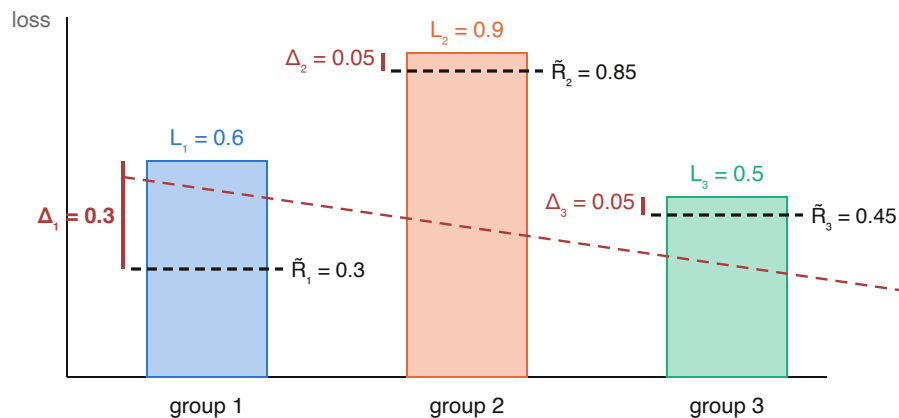
$\psi : Z \rightarrow Y$, serves every group



$f_g = \psi \circ \phi_g$ for every g ,
including groups that share
no measurement (1 and 3)

Which group gets weight: regret, not loss

every N steps: $\lambda_g \leftarrow \lambda_g \cdot \exp(\gamma [\bar{L}_g - \bar{R}_g])$, then renormalise



group DRO on raw loss picks

group 2 (highest loss, already at its floor)

— capacity spent where nothing is left to gain

regret DRO (ours) picks

group 1 (largest gap above its floor)

— the group the model is actually underserving

min player (every step)

$\theta \leftarrow \theta - \eta \nabla_{\theta} \sum_g \lambda_g [(L_g - \bar{R}_g) + \alpha_{align} L^{align}_g] + \alpha_{sep} L_{sep}$

θ = encoders ϕ_g , head ψ , anchors (m_c, Σ_c)

$\bar{R}_g$ is a constant: it moves λ , not the gradient

dashed = estimated floor $\bar{R}_g$ (eq. 13); red = regret $\Delta_g = L_g - \bar{R}_g$, the part a model can still remove